

Hotel Management Application

Application Endpoints

Hotel & Hotel–Amenity Management APIs

Focus: Master data for hotels and hotel-level amenities

Tables: Hotel, Amenity, HotelAmenity

Hotel APIs

- POST /hotels
→ Create a new hotel
- GET /hotels
→ Get all hotels (supports pagination & filtering by location)
- GET /hotels/{hotelId}
→ Get hotel details by ID
- PUT /hotels/{hotelId}
→ Update hotel details
- DELETE /hotels/{hotelId}
→ Soft-delete / deactivate hotel (recommended for enterprise use)

Hotel ↔ Amenity Mapping APIs

- POST /hotels/{hotelId}/amenities/{amenityId}
→ Assign an amenity to a hotel
- DELETE /hotels/{hotelId}/amenities/{amenityId}
→ Remove amenity from hotel
- GET /hotels/{hotelId}/amenities
→ List all amenities available in a hotel

Room Type & Room Inventory APIs

Focus: Room catalog, pricing, and availability

Tables: RoomType, Room

Room Type APIs

- POST /room-types
→ Create room type (pricing, occupancy constraints)
- GET /room-types
→ List all room types
- GET /room-types/{roomTypeId}
→ Get room type details
- PUT /room-types/{roomTypeId}
→ Update price / occupancy / description
- DELETE /room-types/{roomTypeId}
→ Disable room type (enterprise-safe)

Room APIs

- POST /rooms
→ Add room to inventory
- GET /rooms
→ List all rooms (filters: availability, roomType)
- GET /rooms/{roomId}
→ Get room details
- PUT /rooms/{roomId}
→ Update room details
- PATCH /rooms/{roomId}/availability
→ Mark room available / unavailable

Reservation (Booking) APIs

Focus: Core transactional workflow

Tables: Reservation, Room

Enterprise rule: Reservation creation must validate room availability & dates.

Reservation APIs

Hotel Management Application

- POST /reservations
→ Create reservation (check-in / check-out validation)
- GET /reservations
→ List all reservations (date-range search)
- GET /reservations/{reservationId}
→ Get reservation details
- PUT /reservations/{reservationId}
→ Modify reservation dates or customer details
- DELETE /reservations/{reservationId}
→ Cancel reservation

Reservation Lookup APIs

- GET /rooms/{roomId}/reservations
→ View all reservations for a specific room
- GET /reservations/guest-email/{email}
→ Fetch reservations by guest email

Payment APIs

Focus: Payment lifecycle, audit-friendly design

Table: Payment

Enterprise practice: Payments are **append-only**; avoid hard deletes.

Payment APIs

- POST /payments
→ Record payment for a reservation
- GET /payments
→ List all payments (filter by date/status)
- GET /payments/{paymentId}
→ Get payment details
- GET /reservations/{reservationId}/payments
→ Get payment history for a reservation
- PATCH /payments/{paymentId}/status
→ Update payment status (PAID / FAILED / REFUNDED)

Reviews & Room-Amenity APIs

Focus: Customer feedback & room-level amenities

Tables: Review, RoomAmenity, Amenity

Review APIs

- POST /reviews
→ Submit a review (linked to reservation)
- GET /reviews
→ List all reviews
- GET /reservations/{reservationId}/review
→ Get review for a reservation
- GET /hotels/{hotelId}/reviews
→ Aggregated reviews for hotel (via reservations → rooms)
- DELETE /reviews/{reviewId}
→ Remove inappropriate review (admin use)

Room ↔ Amenity APIs

- POST /rooms/{roomId}/amenities/{amenityId}
→ Assign amenity to room
- DELETE /rooms/{roomId}/amenities/{amenityId}
→ Remove amenity from room
- GET /rooms/{roomId}/amenities
→ List amenities available in a room